

Sample End Semester Question Paper - 2025

Subject: Analog and Digital Electronics | Code: EC302 | Semester: 3

Branch: ECE | Regulation: R20 (sample) | Time: 3 Hours | Max Marks:

This is a SAMPLE question paper generated for Backlog Buddy (demo/testing data).
It is not an official university paper and is provided to demonstrate the platform.

PART A - Short questions (10 x 1 = 10 Marks)

1. Name two feedback topologies.
2. What is a Schmitt trigger?
3. Define h-parameters.
4. What is bias stability?
5. What is a differential amplifier?
6. What is a Zener diode?
7. Name two BJT configurations.
8. What is a voltage divider bias?
9. Define small-signal analysis.
10. What is a class-A amplifier?

PART B - Long questions (6 x 13 = 78 Marks)

1. Explain the E-M model of the diode. Draw diode and Zener V-I characteristics.
2. Explain BJT small-signal analysis using h-parameters.
3. Explain biasing techniques: fixed, emitter-stabilised and voltage divider.
4. Explain common-emitter, common-base and common-collector configurations.
5. Explain a class-A amplifier and calculate its efficiency.
6. Explain feedback topologies and their effects on amplifier performance.